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1 Chapter 1: Concentration of sum of independent RV

High dimension probability in Data science

Chapter 1: Concentration of sum of independent RV

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Proposition 1.1 (Tails of the normal distribution). *Let $g \sim N(0, 1)$. Then for all $t > 0$, we have*

$$\left(\frac{1}{t} - \frac{1}{t^3}\right) \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \leq \mathbb{P}\{g \geq t\} \leq \frac{1}{t} \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

In particular, for $t \geq 1$ the tail is bounded by the density:

$$\mathbb{P}\{g \geq t\} \leq \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

Theorem 1.2 (Berry-Esseen central limit theorem).

$$Z_N := \frac{S_N - \mathbb{E}S_N}{\sqrt{\text{Var}(S_N)}} = \frac{1}{\sigma\sqrt{N}} \sum_{i=1}^N (X_i - \mu)$$

for every N and every $t \in \mathbb{R}$ we have

$$|\mathbb{P}\{Z_N \geq t\} - \mathbb{P}\{g \geq t\}| \leq \frac{\rho}{\sqrt{N}}$$

Here $\rho = \mathbb{E}|X_1 - \mu|^3 / \sigma^3$ and $g \sim N(0, 1)$.

Theorem 1.3 (Hoeffding inequality). *Let $X_1, \dots, X_N$ be independent symmetrical Bernoulli random variables, and $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Then, for any $t \geq 0$, we have*

$$\mathbb{P}\left\{\sum_{i=1}^N a_i X_i \geq t\right\} \leq \exp\left(-\frac{t^2}{2\|a\|_2^2}\right)$$

Theorem 1.4 (Hoeffding inequality for general bounded random variable). *Let $X_1, \dots, X_N$ be independent random variables. Assume that $X_i \in [m_i, M_i]$ for every i . Then, for any $t > 0$, we have*

$$\mathbb{P}\left\{\sum_{i=1}^N (X_i - \mathbb{E}X_i) \geq t\right\} \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^N (M_i - m_i)^2}\right)$$

Proposition 1.5 (subgaussian properties).

$$\mathbb{P}\{|X| \geq t\} \leq 2 \exp\left(-ct^2/\|X\|_{\psi_2}^2\right) \quad \text{for all } t \geq 0$$

$$\|X\|_{L^p} \leq C\|X\|_{\psi_2} \sqrt{p} \quad \text{for all } p \geq 1$$

$$\mathbb{E} \exp\left(X^2/\|X\|_{\psi_2}^2\right) \leq 2$$

$$\text{if } \mathbb{E}X = 0 \text{ then } \mathbb{E} \exp(\lambda X) \leq \exp(C\lambda^2\|X\|_{\psi_2}^2) \quad \text{for all } \lambda \in \mathbb{R}.$$

Proposition 1.6 (sum of independent sub-gaussian).

$$\left\|\sum_{i=1}^N X_i\right\|_{\psi_2}^2 \leq C \sum_{i=1}^N \|X_i\|_{\psi_2}^2$$

Theorem 1.7 (General Hoeffding inequality 1). *Let $X_1, \dots, X_N$ be independent, mean zero, sub-gaussian random variables. Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \left| \sum_{i=1}^N X_i \right| \geq t \right\} \leq 2 \exp \left(- \frac{ct^2}{\sum_{i=1}^N \|X_i\|_{\psi_2}^2} \right)$$

Theorem 1.8 (General Hoeffding inequality 2). *Let $X_1, \dots, X_N$ be independent, mean zero, sub-gaussian random variables, and $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \left| \sum_{i=1}^N a_i X_i \right| \geq t \right\} \leq 2 \exp \left(- \frac{ct^2}{K^2 \|a\|_2^2} \right)$$

where $K = \max_i \|X_i\|_{\psi_2}$

Theorem 1.9 (Khintchine's inequality for $p \geq 2$). *Let $X_1, \dots, X_N$ be independent sub-gaussian random variables with zero means and unit variances, and let $a = (a_1, \dots, a_N) \in \mathbb{R}^N$.*

$$\left(\sum_{i=1}^N a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^N a_i X_i \right\|_{L^p} \leq CK \sqrt{p} \left(\sum_{i=1}^N a_i^2 \right)^{1/2}$$

where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

Theorem 1.10 (Khintchine's inequality for $p = 1$). *Same setting as 1.9*

$$c(K) \left(\sum_{i=1}^N a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^N a_i X_i \right\|_{L^1} \leq \left(\sum_{i=1}^N a_i^2 \right)^{1/2}$$

Here $K = \max_i \|X_i\|_{\psi_2}$ and $c(K) > 0$ is a quantity which may depend only on K .

Definition 1.11 (sub-exponential norm).

$$\|X\|_{\psi_1} = \inf \{ t > 0 : \mathbb{E} \exp(|X|/t) \leq 2 \}$$

Lemma 1.12 (Centering). *If X is a sub-gaussian random variable then $X - \mathbb{E}X$ is sub-gaussian, too, and*

$$\begin{aligned} \|X - \mathbb{E}X\|_{\psi_2} &\leq C \|X\|_{\psi_2} \\ \|X - \mathbb{E}X\|_{\psi_1} &\leq C \|X\|_{\psi_1} \end{aligned}$$

where C is an absolute constant.

Lemma 1.13 (sub-exponential is sub-gaussian square). *A random variable X is sub-gaussian if and only if X^2 is sub-exponential. Moreover,*

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$$

Lemma 1.14 (product of sub-gaussian is sub-exponential). *Let X and Y be sub-gaussian random variables. Then XY is sub-exponential. Moreover,*

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}$$

Theorem 1.15 (Bernstein's inequality 1). *Let $X_1, \dots, X_N$ be independent, mean zero, sub-exponential random variables. Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \left| \sum_{i=1}^N X_i \right| \geq t \right\} \leq 2 \exp \left[-c \min \left(\frac{t^2}{\sum_{i=1}^N \|X_i\|_{\psi_1}^2}, \frac{t}{\max_i \|X_i\|_{\psi_1}} \right) \right]$$

where $c > 0$ is an absolute constant.

Remark: The reason we have "min" here is: the bound for MGF does not hold for all lambda in case of subexponential

Theorem 1.16 (Bernstein's inequality 2). *Let $X_1, \dots, X_N$ be independent, mean zero, sub-exponential random variables, and $a = (a_1, \dots, a_N) \in \mathbb{R}^N$. Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \left| \sum_{i=1}^N a_i X_i \right| \geq t \right\} \leq 2 \exp \left[-c \min \left(\frac{t^2}{K^2 \|a\|_2^2}, \frac{t}{K \|a\|_\infty} \right) \right]$$

where $K = \max_i \|X_i\|_{\psi_1}$

Theorem 1.17 (Bernstein's inequality for bounded distributions). *Let $X_1, \dots, X_N$ be independent, mean zero random variables, such that $|X_i| \leq K$ all i . Then, for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \left| \sum_{i=1}^N X_i \right| \geq t \right\} \leq 2 \exp \left(-\frac{t^2/2}{\sigma^2 + Kt/3} \right)$$

Here $\sigma^2 = \sum_{i=1}^N \mathbb{E} X_i^2$ is the variance of the sum.

Theorem 1.18 (bounded difference inequality). *Theorem 2.9.1 (Bounded differences inequality). Let $X_1, \dots, X_N$ be independent random variables. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a measurable function. Assume that the value of $f(x)$ can change by at most $c_i > 0$ under an arbitrary change of a single coordinate of $x \in \mathbb{R}^n$. Then, for any $t > 0$, we have*

$$\mathbb{P}\{f(X) - \mathbb{E}f(X) \geq t\} \leq \exp \left(-\frac{2t^2}{\sum_{i=1}^N c_i^2} \right)$$

where $X = (X_1, \dots, X_n)$

Theorem 1.19 (Bennett's inequality). *Let $X_1, \dots, X_N$ be independent random variables. Assume that $|X_i - \mathbb{E}X_i| \leq K$ almost surely for every i . Then, for any $t > 0$, we have*

$$\mathbb{P} \left\{ \sum_{i=1}^N (X_i - \mathbb{E}X_i) \geq t \right\} \leq \exp \left(-\frac{\sigma^2}{K^2} h \left(\frac{Kt}{\sigma^2} \right) \right)$$

where $\sigma^2 = \sum_{i=1}^N \text{Var}(X_i)$ is the variance of the sum, and $h(u) = (1+u) \log(1+u) - u$.

2 Chapter 2: Random vectors in high dimensions

High dimension probability in Data science

Chapter 2: Random vectors in high dimensions

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Theorem 2.1 (Concentration of norm). *Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, sub-gaussian coordinates X_i that satisfy $\mathbb{E}X_i^2 = 1$. Then*

$$\|X\|_2 - \sqrt{n} \leq CK^2$$

$$\mathbb{P}\{|\|X\|_2 - \sqrt{n}| \geq t\} \leq 2 \exp\left(-\frac{ct^2}{K^4}\right) \quad \text{for all } t \geq 0$$

where $K = \max_i \|X_i\|_{\psi_2}$ and C is an absolute constant.

Lemma 2.2 (Characteristic of isotropy). *A random vector X in $\mathbb{R}^n$ is isotropic if and only if*

$$\mathbb{E}\langle X, x \rangle^2 = \|x\|_2^2 \quad \text{for all } x \in \mathbb{R}^n.$$

Lemma 2.3. *Let X be an isotropic random vector in $\mathbb{R}^n$. Then*

$$\mathbb{E}\|X\|_2^2 = n$$

Moreover, if X and Y are two independent isotropic random vectors in $\mathbb{R}^n$, then

$$\mathbb{E}\langle X, Y \rangle^2 = n$$

Lemma 2.4 (Normal and spherical distributions). *Let us represent $g \sim N(0, I_n)$ in polar form as*

$$g = r\theta$$

where $r = \|g\|_2$ is the length and $\theta = g/\|g\|_2$ is the direction of g . Then

- (a) The length r and direction θ are independent random variables.
- (b) The direction θ is uniformly distributed on the unit sphere S^{n-1} .

Lemma 2.5 (Sub-gaussian distributions with independent coordinates). *Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, subgaussian coordinates X_i . Then X is a sub-gaussian random vector, and*

$$\|X\|_{\psi_2} \leq C \max_{i \leq n} \|X_i\|_{\psi_2}$$

Theorem 2.6 (Uniform distribution on the sphere is sub-gaussian). *Let X be a random vector uniformly distributed on the Euclidean sphere in $\mathbb{R}^n$ with center at the origin and radius $\sqrt{n}$:*

$$X \sim \text{Unif}(\sqrt{n}S^{n-1})$$

Then X is sub-gaussian, and

$$\|X\|_{\psi_2} \leq C$$

Remark: result also hold for Uniform distribution on the Euclidean ball $\text{Unif}(B(0, \sqrt{n}))$.

Theorem 2.7 (Projective limit theorem). *$X \sim \text{Unif}(\sqrt{n}S^{n-1})$ then for any fixed unit vector x , we have $\langle X, x \rangle \rightarrow N(0, 1)$ in distribution as $n \rightarrow \infty$.*

Theorem 2.8 (Grothendieck's inequality). *Consider an $m \times n$ matrix (a_{ij}) of real numbers. Assume that, for any numbers $x_i, y_j \in \{-1, 1\}$, we have*

$$\left| \sum_{i,j} a_{ij} x_i y_j \right| \leq 1$$

Then, for any Hilbert space H and any vectors $u_i, v_j \in H$ satisfying $\|u_i\| = \|v_j\| = 1$, we have

$$\left| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \right| \leq K$$

where $K \leq 1.783$ is an absolute constant.

Definition 2.9 (semidefinite programme). *A semidefinite program is an optimization problem of the following type:*

$$\text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad \langle B_i, X \rangle = b_i \text{ for } i = 1, \dots, m$$

Here A and B_i are given $n \times n$ matrices and b_i are given real numbers.

Theorem 2.10. *Consider two optimization problem: for a given matrix $A \in \mathbb{R}^{n \times n}$*

$$\text{INT}(A) = \text{maximize } \sum_{i,j=1}^n A_{ij} x_i x_j : \quad x_i = \pm 1 \text{ for } i = 1, \dots, n$$

$$\text{SDP}(A) = \text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad X_{ii} = 1 \text{ for } i = 1, \dots, n$$

Then

$$\text{INT}(A) \leq \text{SDP}(A) \leq 2K \cdot \text{INT}(A)$$

where $K \leq 1.783$ is the constant in Grothendieck's inequality.

Theorem 2.11 (Max-cut and SDP relaxation). *Given graph G and adjacency matrix A*

$$\text{MAX-CUT}(G) = \frac{1}{4} \max \left\{ \sum_{i,j=1}^n A_{ij} (1 - x_i x_j) : x_i = \pm 1 \text{ for all } i \right\}$$

$$\text{SDP}(G) := \frac{1}{4} \max \left\{ \sum_{i,j=1}^n A_{ij} (1 - \langle X_i, X_j \rangle) : X_i \in \mathbb{R}^n, \|X_i\|_2 = 1 \text{ for all } i \right\}$$

Let $x = (x_i)$ be the result of a randomized rounding of the solution (X_i) of the semidefinite program, which means that

$$g \sim N(0, I_n) \\ x_i := \text{sign} \langle X_i, g \rangle, \quad i = 1, \dots, n$$

Then we have

$$\mathbb{E} \text{CUT}(G, x) \geq 0.878 \text{SDP}(G) \geq 0.878 \text{MAX-CUT}(G)$$

Lemma 2.12 (Grothendieck's identity). *Consider a random vector $g \sim N(0, I_n)$. Then, for any fixed vectors $u, v \in S^{n-1}$, we have*

$$\mathbb{E} \text{sign} \langle g, u \rangle \text{sign} \langle g, v \rangle = \frac{2}{\pi} \arcsin \langle u, v \rangle$$

3 Chapter 3: Random matrices

High dimension probability in Data science

Chapter 3: Random matrices

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Definition 3.1 (operator norm quadratic form).

$$\|A\| = \max_{x \in S^{n-1}, y \in S^{m-1}} \langle Ax, y \rangle$$

Lemma 3.2 (Approximate isometry). *Let A be an $m \times n$ matrix and $\delta > 0$. Suppose that*

$$\|A^\top A - I_n\| \leq \max(\delta, \delta^2).$$

Then

$$(1 - \delta)\|x\|_2 \leq \|Ax\|_2 \leq (1 + \delta)\|x\|_2 \quad \text{for all } x \in \mathbb{R}^n.$$

Consequently, all singular values of A are between $1 - \delta$ and $1 + \delta$:

$$1 - \delta \leq s_n(A) \leq s_1(A) \leq 1 + \delta.$$

Definition 3.3 (packing number). *A subset $\mathcal{N}$ of a metric space (T, d) is ε -separated if $d(x, y) > \varepsilon$ for all distinct points $x, y \in \mathcal{N}$. The largest possible cardinality of an ε -separated subset of a given set $K \subset T$ is called the packing number of K and is denoted $\mathcal{P}(K, d, \varepsilon)$.*

Lemma 3.4 (Nets from separated sets). *Let $\mathcal{N}$ be a maximal 2ε -separated subset of K . Then $\mathcal{N}$ is an ε -net of K .*

Lemma 3.5 (Equivalence of covering and packing numbers). *For any set $K \subset T$ and any $\varepsilon > 0$, we have*

$$\mathcal{P}(K, d, 2\varepsilon) \leq \mathcal{N}(K, d, \varepsilon) \leq \mathcal{P}(K, d, \varepsilon)$$

Proposition 3.6 (Covering numbers and volume). *Let K be a subset of $\mathbb{R}^n$ and $\varepsilon > 0$. Then*

$$\frac{|K|}{|\varepsilon B_2^n|} \leq \mathcal{N}(K, \varepsilon) \leq \mathcal{P}(K, \varepsilon) \leq \frac{|(K + (\varepsilon/2)B_2^n)|}{|(\varepsilon/2)B_2^n|}$$

Corollary 3.7 (Covering numbers of the Euclidean ball).

$$\left(\frac{1}{\varepsilon}\right)^n \leq \mathcal{N}(B_2^n, \varepsilon) \leq \left(\frac{2}{\varepsilon} + 1\right)^n.$$

The same upper bound is true for the unit Euclidean sphere S^{n-1} .

Lemma 3.8 (Computing the operator norm on a net). *Let A be an $m \times n$ matrix and $\varepsilon \in [0, 1]$. Then, for any ε -net $\mathcal{N}$ of the sphere S^{n-1} , we have*

$$\sup_{x \in \mathcal{N}} \|Ax\|_2 \leq \|A\| \leq \frac{1}{1 - \varepsilon} \cdot \sup_{x \in \mathcal{N}} \|Ax\|_2$$

Theorem 3.9 (Norm of matrices with sub-gaussian entries). *Let A be an $m \times n$ random matrix whose entries A_{ij} are independent, mean zero, sub-gaussian random variables. Then, for any $t > 0$ we have ⁶*

$$\|A\| \leq CK(\sqrt{m} + \sqrt{n} + t)$$

with probability at least $1 - 2 \exp(-t^2)$. Here $K = \max_{i,j} \|A_{ij}\|_{\psi_2}$.

Corollary 3.10 (Norm of symmetric matrices with sub-gaussian entries). *Let A be an $n \times n$ symmetric random matrix whose entries A_{ij} on and above the diagonal are independent, mean zero, sub-gaussian random variables. Then, for any $t > 0$ we have*

$$\|A\| \leq CK(\sqrt{n} + t)$$

with probability at least $1 - 4 \exp(-t^2)$. Here $K = \max_{i,j} \|A_{ij}\|_{\psi_2}$.

Theorem 3.11 (Weyl's inequality). *For any symmetric matrices S and T with the same dimensions, we have*

$$\max_i |\lambda_i(S) - \lambda_i(T)| \leq \|S - T\|$$

Theorem 3.12 (Davis Kahan). *Let S and T be symmetric matrices with the same dimensions. Fix i and assume that the i -th largest eigenvalue of S is well separated from the rest of the spectrum:*

$$\min_{j:j \neq i} |\lambda_i(S) - \lambda_j(S)| = \delta > 0$$

Then the angle between the eigenvectors of S and T corresponding to the i -th largest eigenvalues (as a number between 0 and $\pi/2$) satisfies

$$\sin \angle(v_i(S), v_i(T)) \leq \frac{2\|S - T\|}{\delta}$$

and

$$\exists \theta \in \{-1, 1\} : \quad \|v_i(S) - \theta v_i(T)\|_2 \leq \frac{2^{3/2}\|S - T\|}{\delta}$$

Theorem 3.13 (Two-sided bound on sub-gaussian matrices, not sharp). *Let A be an $m \times n$ matrix whose rows A_i are independent, mean zero, sub-gaussian isotropic random vectors in $\mathbb{R}^n$. Then for any $t \geq 0$ we have*

$$\sqrt{m} - CK^2(\sqrt{n} + t) \leq s_n(A) \leq s_1(A) \leq \sqrt{m} + CK^2(\sqrt{n} + t)$$

with probability at least $1 - 2 \exp(-t^2)$. Here $K = \max_i \|A_i\|_{\psi_2}$.

Theorem 3.14 (covariance estimation). *Let X be a sub-gaussian random vector in $\mathbb{R}^n$. More precisely, assume that there exists $K \geq 1$ such that*

$$\|\langle X, x \rangle\|_{\psi_2} \leq K \|\langle X, x \rangle\|_{L^2} \quad \text{for any } x \in \mathbb{R}^n.$$

Then, for every positive integer m , we have

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq CK^2 \left(\sqrt{\frac{n}{m}} + \frac{n}{m} \right) \|\Sigma\|$$

OR

$$\|\Sigma_m - \Sigma\| \leq CK^2 \left(\sqrt{\frac{n+u}{m}} + \frac{n+u}{m} \right) \|\Sigma\|$$

with probability at least $1 - 2e^{-u}$.

4 Chapter 4: Concentration without independence

High dimension probability in Data science

Chapter 4: Concentration without independence

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Theorem 4.1 (Concentration of Lipschitz functions on the sphere). *Consider a random vector $X \sim \text{Unif}(\sqrt{n}S^{n-1})$, i.e. X is uniformly distributed on the Euclidean sphere of radius $\sqrt{n}$. Consider a Lipschitz function $f : \sqrt{n}S^{n-1} \rightarrow \mathbb{R}$. Then*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq C\|f\|_{\text{Lip}}$$

$$\mathbb{P}\{|f(X) - \mathbb{E}f(X)| \geq t\} \leq 2 \exp\left(-\frac{ct^2}{\|f\|_{\text{Lip}}^2}\right)$$

Theorem 4.2 (Isoperimetric inequality on $\mathbb{R}^n$). *Among all subsets $A \subset \mathbb{R}^n$ with given volume, the Euclidean balls have minimal area. Moreover, for any $\varepsilon > 0$, the Euclidean balls minimize the volume of the ε -neighborhood of A , defined as ²*

$$A_\varepsilon := \{x \in \mathbb{R}^n : \exists y \in A \text{ such that } \|x - y\|_2 \leq \varepsilon\} = A + \varepsilon B_2^n$$

Theorem 4.3 (Isoperimetric inequality on the sphere). *Let $\varepsilon > 0$. Then, among all sets $A \subset S^{n-1}$ with given area $\sigma_{n-1}(A)$, the spherical caps minimize the area of the neighborhood $\sigma_{n-1}(A_\varepsilon)$, where*

$$A_\varepsilon := \{x \in S^{n-1} : \exists y \in A \text{ such that } \|x - y\|_2 \leq \varepsilon\}$$

Lemma 4.4 (blow-up). *Let A be a subset of the sphere $\sqrt{n}S^{n-1}$, and let σ denote the normalized area on that sphere. If $\sigma(A) \geq 1/2$, then, for every $t \geq 0$,*

$$\sigma(A_t) \geq 1 - 2 \exp(-ct^2)$$

Lemma 4.5 (Concentration about expectation and median are equivalent). *Consider a random variable Z with median M . Show that*

$$c\|Z - \mathbb{E}Z\|_{\psi_2} \leq \|Z - M\|_{\psi_2} \leq C\|Z - \mathbb{E}Z\|_{\psi_2}$$

Theorem 4.6 (Gaussian isoperimetric inequality). *Let $\varepsilon > 0$. Then, among all sets $A \subset \mathbb{R}^n$ with fixed Gaussian measure $\gamma_n(A)$, the half spaces minimize the Gaussian measure of the neighborhood $\gamma_n(A_\varepsilon)$.*

Theorem 4.7 (Gaussian concentration). *Consider a random vector $X \sim N(0, I_n)$ and a Lipschitz function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (with respect to the Euclidean metric). Then*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq C\|f\|_{\text{Lip}}$$

Theorem 4.8 (Concentration on the Hamming cube). *Consider a random vector $X \sim \text{Unif}\{0, 1\}^n$. (Thus, the coordinates of X are independent $\text{Ber}(1/2)$ random variables.) Consider a function $f : \{0, 1\}^n \rightarrow \mathbb{R}$. Then*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq \frac{C\|f\|_{\text{Lip}}}{\sqrt{n}}$$

Theorem 4.9 (Concentration on the continuous cube). *Consider a random vector $X \sim \text{Unif}([0, 1]^n)$. (Thus, the coordinates of X are independent random variables uniformly distributed on $[0, 1]$.) Consider a Lipschitz function $f : [0, 1]^n \rightarrow \mathbb{R}$. (The Lipschitz norm is with respect to the Euclidean distance.) Then*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq C\|f\|_{\text{Lip}}$$

Theorem 4.10 (Concentration on the Euclidean ball). *Consider the random vector $X \sim \text{Unif}(\sqrt{n}B_2^n)$. Consider a Lipschitz function $f : \sqrt{n}B_2^n \rightarrow \mathbb{R}$. (The Lipschitz norm is with respect to the Euclidean distance.) Then*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq C\|f\|_{\text{Lip}}$$

Theorem 4.11 (Concentration of concave density). *Consider a random vector X in $\mathbb{R}^n$ whose density has the form $f(x) = e^{-U(x)}$ for some function $U : \mathbb{R}^n \rightarrow \mathbb{R}$. Assume there exists $\kappa > 0$ such that $^{\text{12}}$ $\text{Hess } U(x) \succeq \kappa I_n$ for all $x \in \mathbb{R}^n$. Then any Lipschitz function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies*

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq \frac{C\|f\|_{\text{Lip}}}{\sqrt{\kappa}}$$

Theorem 4.12 (Talagrand's concentration inequality). *Consider a random vector $X = (X_1, \dots, X_n)$ whose coordinates are independent and satisfy*

$$|X_i| \leq 1 \quad \text{almost surely.}$$

Then for any convex Lipschitz function $f : [0, 1]^n \rightarrow \mathbb{R}$

$$\|f(X) - \mathbb{E}f(X)\|_{\psi_2} \leq \frac{C\|f\|_{\text{Lip}}}{\sqrt{\kappa}}$$

Remark: In particular, Talagrand's concentration inequality holds for any norm on $\mathbb{R}^n$.

Theorem 4.13 (Johnson-Lindenstrauss Lemma). *Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$ and $\varepsilon > 0$. Assume that*

$$m \geq (C/\varepsilon^2) \log N$$

Consider a random m -dimensional subspace E in $\mathbb{R}^n$ uniformly distributed in $G_{n,m}$. Denote the orthogonal projection onto E by P . Then, with probability at least $1 - 2\exp(-c\varepsilon^2 m)$, the scaled projection

$$Q := \sqrt{\frac{n}{m}}P$$

is an approximate isometry on $\mathcal{X}$:

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_2 \leq (1 + \varepsilon)\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}$$

Remark: Let A be an $m \times n$ random matrix whose rows are independent, mean zero, subgaussian isotropic random vectors in $\mathbb{R}^n$. Show that the conclusion of JohnsonLindenstrauss lemma holds for $Q = (1/\sqrt{n})A$.

Lemma 4.14 (Random projection). *Let P be a projection in $\mathbb{R}^n$ onto a random m -dimensional subspace uniformly distributed in $G_{n,m}$. Let $z \in \mathbb{R}^n$ be a (fixed) point and $\varepsilon > 0$. Then:*

(a)

$$(\mathbb{E}\|Pz\|_2^2)^{1/2} = \sqrt{\frac{m}{n}}\|z\|_2$$

(b) *With probability at least $1 - 2\exp(-c\varepsilon^2 m)$, we have*

$$(1 - \varepsilon)\sqrt{\frac{m}{n}}\|z\|_2 \leq \|Pz\|_2 \leq (1 + \varepsilon)\sqrt{\frac{m}{n}}\|z\|_2$$

Theorem 4.15 (Matrix Bernstein's inequality). *Let $X_1, \dots, X_N$ be independent, mean zero, $n \times n$ symmetric random matrices, such that $\|X_i\| \leq K$ almost surely for all i . Then, for every $t \geq 0$, we have*

$$\mathbb{P}\left\{\left\|\sum_{i=1}^N X_i\right\| \geq t\right\} \leq 2n \exp\left(-\frac{t^2/2}{\sigma^2 + Kt/3}\right)$$

Here $\sigma^2 = \left\| \sum_{i=1}^N \mathbb{E} X_i^2 \right\|$ is the norm of the matrix variance of the sum. In particular, we can express this bound as the mixture of sub-gaussian and sub-exponential tail, just like in the scalar Bernstein's inequality:

$$\mathbb{P} \left\{ \left\| \sum_{i=1}^N X_i \right\| \geq t \right\} \leq 2n \exp \left[-c \cdot \min \left(\frac{t^2}{\sigma^2}, \frac{t}{K} \right) \right]$$

Remark: $\max(a, b) \asymp a + b$

For rectangular matrix $m \times n$,

$$\mathbb{P} \left\{ \left\| \sum_{i=1}^N X_i \right\| \geq t \right\} \leq 2(m+n) \exp \left(-\frac{t^2/2}{\sigma^2 + Kt/3} \right)$$

where

$$\sigma^2 = \max \left(\left\| \sum_{i=1}^N \mathbb{E} X_i^\top X_i \right\|, \left\| \sum_{i=1}^N \mathbb{E} X_i X_i^\top \right\| \right)$$

Theorem 4.16 (Golden-Thompson inequality). For any $n \times n$ symmetric matrices A and B , we have

$$\text{tr} (e^{A+B}) \leq \text{tr} (e^A e^B)$$

Unfortunately, Golden-Thompson inequality does not hold for three or more matrices: in general, the inequality $\text{tr} (e^{A+B+C}) \leq \text{tr} (e^A e^B e^C)$ may fail.

Theorem 4.17 (Lieb's inequality). Let H be an $n \times n$ symmetric matrix. Define the function on matrices

$$f(X) := \text{tr} \exp(H + \log X)$$

Then f is concave on the space on positive definite $n \times n$ symmetric matrices.

Theorem 4.18 (Lieb's inequality for random matrices). Let H be a fixed $n \times n$ symmetric matrix and Z be a random $n \times n$ symmetric matrix. Then

$$\mathbb{E} \text{tr} \exp(H + Z) \leq \text{tr} \exp(H + \log \mathbb{E} e^Z)$$

Lemma 4.19 (Moment generating function). Let X be an $n \times n$ symmetric mean zero random matrix such that $\|X\| \leq K$ almost surely. Then

$$\mathbb{E} \exp(\lambda X) \preceq \exp(g(\lambda) \mathbb{E} X^2) \quad \text{where} \quad g(\lambda) = \frac{\lambda^2/2}{1 - |\lambda|K/3}$$

provided that $|\lambda| < 3/K$.

Theorem 4.20 (Matrix Bernstein's inequality: expectation). Let $X_1, \dots, X_N$ be independent, mean zero, $n \times n$ symmetric random matrices, such that $\|X_i\| \leq K$ almost surely for all i . Deduce from Bernstein's inequality that

$$\mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \lesssim \left\| \sum_{i=1}^N \mathbb{E} X_i^2 \right\|^{1/2} \sqrt{1 + \log n} + K(1 + \log n).$$

Theorem 4.21 (Matrix Hoeffding's inequality). Let $\varepsilon_1, \dots, \varepsilon_n$ be independent symmetric Bernoulli random variables and let $A_1, \dots, A_N$ be symmetric $n \times n$ matrices (deterministic). Prove that, for any $t \geq 0$, we have

$$\mathbb{P} \left\{ \left\| \sum_{i=1}^N \varepsilon_i A_i \right\| \geq t \right\} \leq 2n \exp(-t^2/2\sigma^2)$$

where $\sigma^2 = \left\| \sum_{i=1}^N A_i^2 \right\|$.

Theorem 4.22 (Matrix Khintchine's inequality). *Let $\varepsilon_1, \dots, \varepsilon_N$ be independent symmetric Bernoulli random variables and let $A_1, \dots, A_N$ be symmetric $n \times n$ matrices (deterministic).*

$$\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i A_i \right\| \leq C \sqrt{1 + \log n} \left\| \sum_{i=1}^N A_i^2 \right\|^{1/2}$$

More generally, prove that for every $p \in [1, \infty)$ we have

$$\left(\mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i A_i \right\|^p \right)^{1/p} \leq C \sqrt{p + \log n} \left\| \sum_{i=1}^N A_i^2 \right\|^{1/2}$$

Theorem 4.23 (General covariance estimation). *Let X be a random vector in $\mathbb{R}^n$, $n \geq 2$. Assume that for some $K \geq 1$,*

$$\|X\|_2 \leq K (\mathbb{E} \|X\|_2^2)^{1/2} \quad \text{almost surely}$$

Then, for every positive integer m , we have

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq C \left(\sqrt{\frac{K^2 n \log n}{m}} + \frac{K^2 n \log n}{m} \right) \|\Sigma\|$$

Theorem 4.24 (Low dimension covariance estimation). *Intrinsic dimension $r = \frac{\text{tr}(\Sigma)}{\|\Sigma\|}$*

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq C \left(\sqrt{\frac{K^2 r \log n}{m}} + \frac{K^2 r \log n}{m} \right) \|\Sigma\|$$

In particular, this stronger bound implies that a sample of size

$$m \asymp \varepsilon^{-2} r \log n$$

is sufficient to estimate the covariance matrix.

Tail bound:

$$\|\Sigma_m - \Sigma\| \leq C \left(\sqrt{\frac{K^2 r (\log n + u)}{m}} + \frac{K^2 r (\log n + u)}{m} \right) \|\Sigma\|$$

with probability at least $1 - 2e^{-u}$. Here $r = \text{tr}(\Sigma)/\|\Sigma\| \leq n$.

5 Chapter 5: Quadratic forms, symmetrization and contraction

High dimension probability in Data science

Chapter 5: Quadratic forms, symmetrization and contraction

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Lemma 5.1. *Let Y and Z be independent random variables such that $\mathbb{E}Z = 0$. Then, for every convex function F , one has*

$$\mathbb{E}F(Y) \leq \mathbb{E}F(Y + Z)$$

Theorem 5.2 (Decoupling). *Let A be an $n \times n$, diagonal-free matrix (i.e. the diagonal entries of A equal zero). Let $X = (X_1, \dots, X_n)$ be a random vector with independent mean zero coordinates X_i . Then, for every convex function $F : \mathbb{R} \rightarrow \mathbb{R}$, one has*

$$\mathbb{E}F(X^\top A X) \leq \mathbb{E}F(4X^\top A X')$$

where X' is an independent copy of X .

Stronger version: for any square matrix $A = (a_{ij})$ we have

$$\mathbb{E}F\left(\sum_{i,j:i \neq j} a_{ij} X_i X_j\right) \leq \mathbb{E}F\left(4 \sum_{i,j} a_{ij} X_i X'_j\right)$$

Theorem 5.3 (Hanson-Wright inequality). *Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, sub-gaussian coordinates. Let A be an $n \times n$ matrix. Then, for every $t \geq 0$, we have*

$$\mathbb{P}\{|X^\top A X - \mathbb{E}X^\top A X| \geq t\} \leq 2 \exp\left[-c \min\left(\frac{t^2}{K^4 \|A\|_F^2}, \frac{t}{K^2 \|A\|}\right)\right]$$

where $K = \max_i \|X_i\|_{\psi_2}$

Lemma 5.4 (MGF of Gaussian chaos). *Let $X, X' \sim N(0, I_n)$ be independent and let $A = (a_{ij})$ be an $n \times n$ matrix. Then*

$$\mathbb{E} \exp(\lambda X^\top A X') \leq \exp(C \lambda^2 \|A\|_F^2)$$

for all λ satisfying $|\lambda| \leq c/\|A\|$.

Lemma 5.5 (Comparison). *Consider independent, mean zero, sub-gaussian random vectors X, X' in $\mathbb{R}^n$ with $\|X\|_{\psi_2} \leq K$ and $\|X'\|_{\psi_2} \leq K$. Consider also independent random vectors $g, g' \sim N(0, I_n)$. Let A be an $n \times n$ matrix. Then*

$$\mathbb{E} \exp(\lambda X^\top A X') \leq \mathbb{E} \exp(C K^2 \lambda g^\top A g')$$

for any $\lambda \in \mathbb{R}$.

Theorem 5.6 (Higher-dimensional Hanson-Wright inequality). *Let $X_1, \dots, X_n$ be independent, mean zero, sub-gaussian random vectors in $\mathbb{R}^d$. Let $A = (a_{ij})$ be an $n \times n$ matrix. Prove that for every $t \geq 0$, we have*

$$\mathbb{P}\left\{\left|\sum_{i,j:i \neq j} a_{ij} \langle X_i, X_j \rangle\right| \geq t\right\} \leq 2 \exp\left[-c \min\left(\frac{t^2}{K^4 d \|A\|_F^2}, \frac{t}{K^2 \|A\|}\right)\right]$$

where $K = \max_i \|X_i\|_{\psi_2}$.

Theorem 5.7 (Concentration of random vectors). *Let B be an $m \times n$ matrix, and let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ be a random vector with independent, mean zero, unit variance, sub-gaussian coordinates. Then*

$$\| \|BX\|_2 - \|B\|_F \|X\|_{\psi_2} \| \leq CK^2 \|B\|$$

where $K = \max_i \|X_i\|_{\psi_2}$

Lemma 5.8 (Symmetrization). *Let $X_1, \dots, X_N$ be independent, mean zero random vectors in a normed space. Then*

$$\frac{1}{2} \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 2 \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\|.$$

Lemma 5.9 (Symmetrization general). *Let $F : \mathbb{R}_+ \rightarrow \mathbb{R}$ be an increasing, convex function. Show that the same inequalities in Lemma 5.8 hold if the norm $\|\cdot\|$ is replaced with $F(\|\cdot\|)$, namely*

$$\mathbb{E} F \left(\frac{1}{2} \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right) \leq \mathbb{E} F \left(\left\| \sum_{i=1}^N X_i \right\| \right) \leq \mathbb{E} F \left(2 \left\| \sum_{i=1}^N \varepsilon_i X_i \right\| \right)$$

Theorem 5.10 (Norms of random matrices with non-i.i.d. entries). *Let A be an $n \times n$ symmetric random matrix whose entries on and above the diagonal are independent, mean zero random variables. Then*

$$\mathbb{E} \|A\| \leq C \sqrt{\log n} \cdot \mathbb{E} \max_i \|A_i\|_2$$

where A_i denote the rows of A .

*Remark: do not confuse with norm for gaussian matrix, this is a general result, hence not sharp.
For rectangular matrix:*

$$\mathbb{E} \|A\| \leq C \sqrt{\log(m+n)} \left(\mathbb{E} \max_i \|A_i\|_2 + \mathbb{E} \max_j \|A^j\|_2 \right)$$

where A_i and A^j denote the rows and columns of A , respectively.

Theorem 5.11 (Matrix completion). $\text{Rank}(X) = r, r \ll n, p = m/n^2$

$$Y_{ij} := \delta_{ij} X_{ij} \quad \text{where} \quad \delta_{ij} \sim \text{Ber}(p) \text{ are independent.}$$

we are shown m entries of X on average. Let $\hat{X}$ be a best rank r approximation to $p^{-1}Y$. Then

$$\mathbb{E} \frac{1}{n} \|\hat{X} - X\|_F \leq C \sqrt{\frac{rn \log n}{m}} \|X\|_\infty$$

as long as $m \geq n \log n$. Here $\|X\|_\infty = \max_{i,j} |X_{ij}|$ is the maximum magnitude of the entries of X .

Theorem 5.12 (Contraction Principle). *Let $\varepsilon_1, \varepsilon_2, \varepsilon_3, \dots$ a sequence of independent symmetric Bernoulli random variables, $x_1, \dots, x_N$ be (deterministic) vectors in some normed space, and let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. Then*

$$\mathbb{E} \left\| \sum_{i=1}^N a_i \varepsilon_i x_i \right\| \leq \|a\|_\infty \cdot \mathbb{E} \left\| \sum_{i=1}^N \varepsilon_i x_i \right\|$$

Lemma 5.13 (Symmetrization with Gaussians). *Let $X_1, \dots, X_N$ be independent, mean zero random vectors in a normed space. Let $g_1, \dots, g_N \sim N(0, 1)$ be independent Gaussian random variables, which are also independent of X_i . Then*

$$\frac{c}{\sqrt{\log N}} \mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\| \leq \mathbb{E} \left\| \sum_{i=1}^N X_i \right\| \leq 3 \mathbb{E} \left\| \sum_{i=1}^N g_i X_i \right\|.$$

Lemma 5.14 (Talagrand's contraction principle). *Consider a bounded subset $T \subset \mathbb{R}^n$, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent symmetric Bernoulli random variables. Let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be contractions, i.e. Lipschitz functions with $\|\phi_i\|_{Lip} \leq 1$. Then*

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^n \varepsilon_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^n \varepsilon_i t_i$$

Lemma 5.15 (Gaussian contraction principle). *Consider a bounded subset $T \subset \mathbb{R}^n$, and let $g_1, \dots, g_n$ be independent $N(0, 1)$. Let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be contractions, i.e. Lipschitz functions with $\|\phi_i\|_{Lip} \leq 1$. Then*

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i t_i$$

6 Chapter 6: Random processes

High dimension probability in Data science

Chapter 6: Random processes

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Definition 6.1. *Increments of the random process are defined as*

$$d(t, s) := \|X_t - X_s\|_{L^2} = \left(\mathbb{E} (X_t - X_s)^2 \right)^{1/2}, \quad t, s \in T$$

Lemma 6.2 (Symmetrization for random processes). *Let $X_1(t), \dots, X_N(t)$ be N independent, mean zero random processes indexed by points $t \in T$. Let $\varepsilon_1, \dots, \varepsilon_N$ be independent symmetric Bernoulli random variables. Prove that*

$$\frac{1}{2} \mathbb{E} \sup_{t \in T} \sum_{i=1}^N \varepsilon_i X_i(t) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^N X_i(t) \leq 2 \mathbb{E} \sup_{t \in T} \sum_{i=1}^N \varepsilon_i X_i(t)$$

Theorem 6.3 (Slepian's inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean zero Gaussian processes. Assume that for all $t, s \in T$, we have*

$$\mathbb{E} X_t^2 = \mathbb{E} Y_t^2 \quad \text{and} \quad \mathbb{E} (X_t - X_s)^2 \leq \mathbb{E} (Y_t - Y_s)^2$$

Then for every $\tau \in \mathbb{R}$ we have

$$\mathbb{P} \left\{ \sup_{t \in T} X_t \geq \tau \right\} \leq \mathbb{P} \left\{ \sup_{t \in T} Y_t \geq \tau \right\}$$

Consequently,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t$$

Whenever the tail comparison inequality (7.3) holds, we say that the random variable X is stochastically dominated by the random variable Y .

Lemma 6.4 (Stein identity). *• If $X \sim N(0, \sigma^2)$, for any differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we have*

$$\mathbb{E} X f(X) = \sigma^2 \mathbb{E} f'(X)$$

• Multivariate stein identity: Let $X \sim N(0, \Sigma)$. Then for any differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ we have

$$\begin{aligned} \mathbb{E} X f(X) &= \Sigma \cdot \mathbb{E} \nabla f(X) \\ \iff \mathbb{E} X_i f(X) &= \sum_{j=1}^n \Sigma_{ij} \mathbb{E} \frac{\partial f}{\partial x_j}(X), \quad i = 1, \dots, n \end{aligned}$$

Lemma 6.5 (Gaussian interpolation). *Consider two independent Gaussian random vectors $X \sim N(0, \Sigma^X)$ and $Y \sim N(0, \Sigma^Y)$. Define the interpolation Gaussian vector*

$$Z(u) := \sqrt{u}X + \sqrt{1-u}Y, \quad u \in [0, 1]$$

Then for any twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E} \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(Z(u)) \right]$$

Lemma 6.6 (Slepian's inequality, functional form). *Consider two mean zero Gaussian random vectors X and Y in $\mathbb{R}^n$. Assume that for all $i, j = 1, \dots, n$, we have*

$$\mathbb{E}X_i^2 = \mathbb{E}Y_i^2 \quad \text{and} \quad \mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2$$

Consider a twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\frac{\partial^2 f}{\partial x_i \partial x_j} \geq 0 \quad \text{for all } i \neq j$$

Then

$$\mathbb{E}f(X) \geq \mathbb{E}f(Y)$$

Theorem 6.7 (Sudakov-Fernique's inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean zero Gaussian processes. Assume that for all $t, s \in T$, we have*

$$\mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t$$

Theorem 6.8 (Gordon's inequality). *Let $(X_{ut})_{u \in U, t \in T}$ and $Y = (Y_{ut})_{u \in U, t \in T}$ be two mean zero Gaussian processes indexed by pairs of points (u, t) in a product set $U \times T$. Assume that we have*

$$\mathbb{E}X_{ut}^2 = \mathbb{E}Y_{ut}^2, \quad \mathbb{E}(X_{ut} - X_{us})^2 \leq \mathbb{E}(Y_{ut} - Y_{us})^2 \quad \text{for all } u, t, s$$

$\mathbb{E}(X_{ut} - X_{vs})^2 \geq \mathbb{E}(Y_{ut} - Y_{vs})^2$ for all $u \neq v$ and all t, s . Then for every $\tau \geq 0$ we have

$$\mathbb{P} \left\{ \inf_{u \in U} \sup_{t \in T} X_{ut} \geq \tau \right\} \leq \mathbb{P} \left\{ \inf_{u \in U} \sup_{t \in T} Y_{ut} \geq \tau \right\}$$

Consequently,

$$\mathbb{E} \inf_{u \in U} \sup_{t \in T} X_{ut} \leq \mathbb{E} \inf_{u \in U} \sup_{t \in T} Y_{ut}$$

Theorem 6.9 (Norms of Gaussian random matrices). *Let A be an $m \times n$ matrix with independent $N(0, 1)$ entries. Then*

$$\begin{aligned} \mathbb{E}\|A\| &\leq \sqrt{m} + \sqrt{n} \\ \mathbb{P}\{\|A\| \geq \sqrt{m} + \sqrt{n} + t\} &\leq 2 \exp(-ct^2) \\ \mathbb{E}s_n(A) &\geq \sqrt{m} - \sqrt{n} \end{aligned}$$

Lemma 6.10 (Symmetric random matrix). *Gaussian orthogonal ensemble (GOE): diagonal entries are independent $N(0, 2)$ random variables.*

$$\mathbb{E}\|A\| \leq 2\sqrt{n}$$

tail bound

$$\mathbb{P}\{\|A\| \geq 2\sqrt{n} + t\} \leq 2 \exp(-ct^2)$$

Theorem 6.11 (Sudakov's minoration inequality). *Let $(X_t)_{t \in T}$ be a mean zero Gaussian process. Then, for any $\varepsilon \geq 0$, we have*

$$\mathbb{E} \sup_{t \in T} X_t \geq c\varepsilon \sqrt{\log \mathcal{N}(T, d, \varepsilon)}$$

where $d(t, s) := \|X_t - X_s\|_{L^2} = \left(\mathbb{E}(X_t - X_s)^2 \right)^{1/2}$.

Corollary 6.12 (Sudakov's minoration inequality in $\mathbb{R}^n$). *Let $T \subset \mathbb{R}^n$. Then, for any $\varepsilon > 0$, we have*

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle \geq c\varepsilon \sqrt{\log \mathcal{N}(T, \varepsilon)}$$

Here $\mathcal{N}(T, \varepsilon)$ is the covering number of T by Euclidean balls – the smallest number of Euclidean balls with radii ε and centers in T that cover T .

Theorem 6.13. $x_1, \dots, x_N$ denote the vertices of P

$$\mathbb{E} \sup_{t \in P} \langle g, t \rangle = \mathbb{E} \sup_{i \leq N} \langle g, x_i \rangle$$

The equality here follows since the maximum of the linear function on the convex set P is attained at an extreme point, i.e. at a vertex of P .

Proposition 6.14 (Gaussian width). • $w(T)$ is finite if and only if T is bounded.

- Gaussian width is invariant under affine unitary transformations. Thus, for every orthogonal matrix U and any vector y , we have

$$w(UT + y) = w(T)$$

- Gaussian width is invariant under taking convex hulls. Thus,

$$w(\text{conv}(T)) = w(T)$$

- Gaussian width respects Minkowski addition of sets and scaling. Thus, for $T, S \subset \mathbb{R}^n$ and $a \in \mathbb{R}$ we have

$$w(T + S) = w(T) + w(S); \quad w(aT) = |a|w(T)$$

- We have

$$w(T) = \frac{1}{2}w(T - T) = \frac{1}{2}\mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle$$

- (Gaussian width and diameter). We have

$$\frac{1}{\sqrt{2\pi}} \cdot \text{diam}(T) \leq w(T) \leq \frac{\sqrt{n}}{2} \cdot \text{diam}(T)$$

- Gaussian width under linear transformations

•

$$w(AT) \leq \|A\|w(T)$$

Definition 6.15 (Spherical width). The spherical width of a subset $T \subset \mathbb{R}^n$ is defined as

$$w_s(T) := \mathbb{E} \sup_{x \in T} \langle \theta, x \rangle \quad \text{where} \quad \theta \sim \text{Unif}(S^{n-1})$$

Lemma 6.16 (Gaussian vs. spherical widths). We have

$$(\sqrt{n} - C)w_s(T) \leq w(T) \leq (\sqrt{n} + C)w_s(T)$$

Definition 6.17 (Squared gaussian width).

$$h(T)^2 := \mathbb{E} \sup_{t \in T} \langle g, t \rangle^2, \quad \text{where} \quad g \sim N(0, I_n)$$

Lemma 6.18.

$$w(T - T) \leq h(T - T) \leq w(T - T) + C_1 \text{diam}(T) \leq Cw(T - T)$$

In particular, we have

$$2w(T) \leq h(T - T) \leq 2Cw(T)$$

Definition 6.19 (stable dimension).

$$d(T) := \frac{h(T - T)^2}{\text{diam}(T)^2} \asymp \frac{w(T)^2}{\text{diam}(T)^2}$$

The stable dimension is always bounded by the algebraic dimension: $d(T) \leq \dim(T)$

Definition 6.20 (Stable rank).

$$r(A) := \frac{\|A\|_F^2}{\|A\|^2}$$

Theorem 6.21 (Sizes of random projections of sets). *Consider a bounded set $T \subset \mathbb{R}^n$. Let P be a projection in $\mathbb{R}^n$ onto a random m -dimensional subspace $E \sim \text{Unif}(G_{n,m})$. Then, with probability at least $1 - 2e^{-m}$, we have*

$$\text{diam}(PT) \leq C \left[w_s(T) + \sqrt{\frac{m}{n}} \text{diam}(T) \right].$$

Optimal:

$$\mathbb{E} \text{diam}(PT) \geq c \left[w_s(T) + \sqrt{\frac{m}{n}} \text{diam}(T) \right]$$

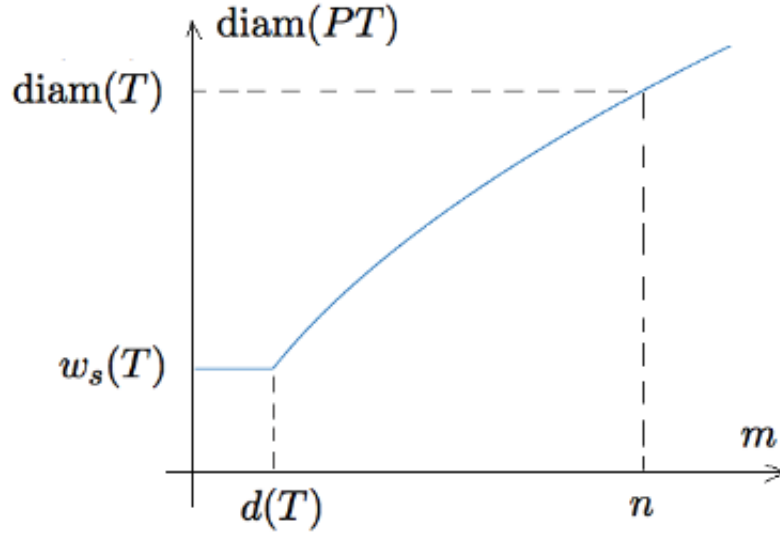


Figure 1: The diameter of a random m -dimensional projection of a set T as a function of m .

7 Chapter 7: Chaining

High dimension probability in Data science

Chapter 7: Chaining

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Definition 7.1 (Sub-gaussian increment). Consider a random process $(X_t)_{t \in T}$ on a metric space (T, d) . We say that the process has sub-gaussian increments if there exists $K \geq 0$ such that

$$\|X_t - X_s\|_{\psi_2} \leq Kd(t, s) \quad \text{for all } t, s \in T$$

where $d(t, s) := \|X_t - X_s\|_{L^2}$, $t, s \in T$

Theorem 7.2 (Dudley's integral inequality). Let $(X_t)_{t \in T}$ be a mean zero random process on a metric space (T, d) with sub-gaussian increments defined above. Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon$$

Remark:

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \int_0^{\text{diam}(T)} \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon$$

Theorem 7.3 (Discrete Dudley's inequality). Let $(X_t)_{t \in T}$ be a mean zero random process on a metric space (T, d) with sub-gaussian increments defined above. Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \sum_{k \in \mathbb{Z}} 2^{-k} \sqrt{\log \mathcal{N}(T, d, 2^{-k})}$$

Theorem 7.4 (Dudley's integral inequality: tail bound). Let $(X_t)_{t \in T}$ be a random process on a metric space (T, d) with sub-gaussian increments as in (8.1). Then, for every $u \geq 0$, the event

$$\sup_{t, s \in T} |X_t - X_s| \leq CK \left[\int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon + u \cdot \text{diam}(T) \right]$$

holds with probability at least $1 - 2 \exp(-u^2)$.

Theorem 7.5 (Dudley's inequality for sets in $\mathbb{R}^n$). For any set $T \subset \mathbb{R}^n$, we have

$$w(T) \leq C \int_0^\infty \sqrt{\log \mathcal{N}(T, \varepsilon)} d\varepsilon$$

For example:

$$w(B_2^n) \leq C \int_0^1 \sqrt{n \log \frac{3}{\varepsilon}} d\varepsilon \leq C_1 \sqrt{n}$$

Theorem 7.6 (Uniform law of large number). $\mathcal{F} := \{f : [0, 1] \rightarrow \mathbb{R}, \|f\|_{Lip} \leq L\}$. Let $X, X_1, X_2, \dots, X_n$ be i.i.d. random variables taking values in $[0, 1]$. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq \frac{CL}{\sqrt{n}}$$

Definition 7.7 (VC dimension). Consider a class $\mathcal{F}$ of Boolean functions on some domain Ω . We say that a subset $\Lambda \subseteq \Omega$ is shattered by $\mathcal{F}$ if any function $g : \Lambda \rightarrow \{0, 1\}$ can be obtained by restricting some function $f \in \mathcal{F}$ onto Λ . The VC dimension of $\mathcal{F}$, denoted $\text{vc}(\mathcal{F})$, is the largest cardinality¹ of a subset $\Lambda \subseteq \Omega$ shattered by $\mathcal{F}$.

Lemma 7.8 (Pajor's Lemma). Let $\mathcal{F}$ be a class of Boolean functions on a finite set Ω . Then

$$|\mathcal{F}| \leq |\{\Lambda \subseteq \Omega : \Lambda \text{ is shattered by } \mathcal{F}\}|$$

We include the empty set $\Lambda = \emptyset$ in the counting on the right side.

Theorem 7.9 (Sauer-Shelah Lemma). Let $\mathcal{F}$ be a class of Boolean functions on an n -point set Ω . Then

$$|\mathcal{F}| \leq \sum_{k=0}^d \binom{n}{k} \leq \left(\frac{en}{d}\right)^d$$

where $d = \text{vc}(\mathcal{F})$.

Definition 7.10 (distance on probability measure μ).

$$d(f, g) = \|f - g\|_{L^2(\mu)} = \left(\int_{\Omega} |f - g|^2 d\mu \right)^{1/2}, \quad f, g \in \mathcal{F}$$

Theorem 7.11 (Covering numbers via VC dimension). Let $\mathcal{F}$ be a class of Boolean functions on a probability space (Ω, Σ, μ) . Then, for every $\varepsilon \in (0, 1)$, we have

$$\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) \leq \left(\frac{2}{\varepsilon}\right)^{Cd}$$

Lemma 7.12 (Dimension reduction). Let $\mathcal{F}$ be a class of N Boolean functions on a probability space (Ω, Σ, μ) . Assume that all functions in $\mathcal{F}$ are ε -separated, that is

$$\|f - g\|_{L^2(\mu)} > \varepsilon \text{ for all distinct } f, g \in \mathcal{F}.$$

Then there exist a number $n \leq C\varepsilon^{-4} \log N$ and an n -point subset $\Omega_n \subset \Omega$ such that the uniform probability measure μ_n on Ω_n satisfies

$$\|f - g\|_{L^2(\mu_n)} > \frac{\varepsilon}{2} \quad \text{for all distinct } f, g \in \mathcal{F}.$$

Theorem 7.13 (Empirical processes via VC dimension). Let $\mathcal{F}$ be a class of Boolean functions on a probability space (Ω, Σ, μ) with finite VC dimension $\text{vc}(\mathcal{F}) \geq 1$. Let $X, X_1, X_2, \dots, X_n$ be independent random points in Ω distributed according to the law μ . Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}f(X) \right| \leq C \sqrt{\frac{\text{vc}(\mathcal{F})}{n}}$$

Lemma 7.14 (Symmetrization for empirical processes). Let $\mathcal{F}$ be a class of functions on a probability space (Ω, Σ, μ) . Let $X, X_1, X_2, \dots, X_n$ be random points in Ω distributed according to the law μ . Prove that

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}f(X) \right| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|$$

where $\varepsilon_1, \varepsilon_2, \dots$ are independent symmetric Bernoulli random variables (which are also independent of $X_1, X_2, \dots$).

Theorem 7.15 (Glivenko-Cantelli Theorem: non-asymptotic). *Let $X_1, \dots, X_n$ be independent random variables with common cumulative distribution function F . Then*

$$\mathbb{E} \|F_n - F\|_\infty = \mathbb{E} \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq \frac{C}{\sqrt{n}}$$

Definition 7.16. *Ideally, we would like to find a function f^* from the hypothesis space $\mathcal{F}$ which would minimize the risk $R(f) = \mathbb{E}(f(X) - T(X))^2$, that is*

$$f^* := \arg \min_{f \in \mathcal{F}} R(f)$$

The empirical risk for a function $f : \Omega \rightarrow \mathbb{R}$ is defined as

$$R_n(f) := \frac{1}{n} \sum_{i=1}^n (f(X_i) - T(X_i))^2.$$

$$f_n^* := \arg \min_{f \in \mathcal{F}} R_n(f)$$

The main question is: how large is the excess risk:

$$R(f_n^*) - R(f^*)$$

Theorem 7.17 (Excess risk via VC dimension). *Assume that the target T is a Boolean function, and the hypothesis space $\mathcal{F}$ is a class of Boolean functions with finite VC dimension $\text{vc}(\mathcal{F}) \geq 1$. Then*

$$\mathbb{E} R(f_n^*) \leq R(f^*) + C \sqrt{\frac{\text{vc}(\mathcal{F})}{n}}$$

Lemma 7.18 (Excess risk via uniform deviations). *We have*

$$R(f_n^*) - R(f^*) \leq 2 \sup_{f \in \mathcal{F}} |R_n(f) - R(f)|$$

pointwise.

Theorem 7.19 (Learning in the class of Lipschitz functions). *Consider*

$$\mathcal{F} := \{f : [0, 1] \rightarrow \mathbb{R}, \|f\|_{\text{Lip}} \leq L\}$$

and a target function $T : [0, 1] \rightarrow [0, 1]$.

- *random process $X_f := R_n(f) - R(f)$ has sub-gaussian increment*

$$\|X_f - X_g\|_{\psi_2} \leq \frac{CL}{\sqrt{n}} \|f - g\|_\infty \quad \text{for all } f, g \in \mathcal{F}$$

•

$$\mathbb{E} \sup_{f \in \mathcal{F}} |R_n(f) - R(f)| \leq \frac{C(L+1)}{\sqrt{n}}$$

•

$$R(f_n^*) - R(f^*) \leq \frac{C(L+1)}{\sqrt{n}}$$

Definition 7.20. $(T_k)_{k=0}^\infty$ *is called an admissible sequence if*

$$|T_0| = 1, \quad |T_k| \leq 2^{2^k}, \quad k = 1, 2, \dots$$

Definition 7.21 (Talagrand's γ_2 functional). Let (T, d) be a metric space. The γ_2 functional of T is defined as

$$\gamma_2(T, d) = \inf_{(T_k)} \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k)$$

where the infimum is with respect to all admissible sequences.

Theorem 7.22 (Generic chaining bound). Let $(X_t)_{t \in T}$ be a mean zero random process on a metric space (T, d) with sub-gaussian increments. Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \gamma_2(T, d)$$

Theorem 7.23 (Generic chaining: tail bound). Let $(X_t)_{t \in T}$ be a random process on a metric space (T, d) with sub-gaussian increments. Then, for every $u \geq 0$, the event

$$\sup_{t, s \in T} |X_t - X_s| \leq CK [\gamma_2(T, d) + u \cdot \text{diam}(T)]$$

holds with probability at least $1 - 2 \exp(-u^2)$.

Theorem 7.24 (Talagrand's majorizing measure theorem). Let $(X_t)_{t \in T}$ be a mean zero Gaussian process on a set T . Consider the canonical metric defined on T by (7.13), i.e. $d(t, s) = \|X_t - X_s\|_{L^2}$. Then

$$c \cdot \gamma_2(T, d) \leq \mathbb{E} \sup_{t \in T} X_t \leq C \cdot \gamma_2(T, d)$$

Corollary 7.25 (Talagrand's comparison inequality). Let $(X_t)_{t \in T}$ be a mean zero random process on a set T and let $(Y_t)_{t \in T}$ be a mean zero Gaussian process. Assume that for all $t, s \in T$, we have

$$\|X_t - X_s\|_{\psi_2} \leq K \|Y_t - Y_s\|_{L^2}$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \mathbb{E} \sup_{t \in T} Y_t$$

Corollary 7.26 (Talagrand's comparison inequality: geometric form). Let $(X_x)_{x \in T}$ be a mean zero random process on a subset $T \subset \mathbb{R}^n$. Assume that for all $x, y \in T$, we have

$$\|X_x - X_y\|_{\psi_2} \leq K \|x - y\|_2$$

Then

$$\mathbb{E} \sup_{x \in T} X_x \leq CK w(T)$$

Theorem 7.27 (Sub-gaussian Chevet's inequality). Let A be an $m \times n$ random matrix whose entries A_{ij} are independent, mean zero, sub-gaussian random variables. Let $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$ be arbitrary bounded sets. Then

$$\mathbb{E} \sup_{x \in T, y \in S} \langle Ax, y \rangle \leq CK [w(T) \text{rad}(S) + w(S) \text{rad}(T)]$$

where $K = \max_{ij} \|A_{ij}\|_{\psi_2}$.

8 Chapter 8: Deviations of random matrices and geometric consequences

High dimension probability in Data science

Ch 8: Deviations of random matrices and geometric consequences

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Theorem 8.1 (Matrix deviation inequality). *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then for any subset $T \subset \mathbb{R}^n$, we have*

$$\mathbb{E} \sup_{x \in T} |\|Ax\|_2 - \sqrt{m}\|x\|_2| \leq CK^2 \gamma(T)$$

Here $\gamma(T)$ is the Gaussian complexity, and $K = \max_i \|A_i\|_{\psi_2}$

Theorem 8.2 (Sub-gaussian increments). *Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random process*

$$X_x := \|Ax\|_2 - \sqrt{m}\|x\|_2$$

has sub-gaussian increments, namely

$$\|X_x - X_y\|_{\psi_2} \leq CK^2 \|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n.$$

Here $K = \max_i \|A_i\|_{\psi_2}$

Lemma 8.3. *Let $x, y \in S^{n-1}$. Then*

$$|\|Ax\|_2 - \|Ay\|_2|_{\psi_2} \leq CK^2 \|x - y\|_2$$

Proposition 8.4 (Sizes of random projections of sets). *Consider a bounded set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the scaled matrix*

$$P := \frac{1}{\sqrt{n}} A$$

(a "sub-gaussian projection") satisfies

$$\mathbb{E} \text{diam}(PT) \leq \sqrt{\frac{m}{n}} \text{diam}(T) + CK^2 w_s(T)$$

Theorem 8.5 (Covariance estimation for lower-dimensional distributions). *Let X be a sub-gaussian random vector in $\mathbb{R}^n$. More precisely, assume that there exists $K \geq 1$ such that*

$$\|\langle X, x \rangle\|_{\psi_2} \leq K \|\langle X, x \rangle\|_{L^2} \quad \text{for any } x \in \mathbb{R}^n.$$

Then, for every positive integer m , we have

$$\mathbb{E} \|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r}{m}} + \frac{r}{m} \right) \|\Sigma\|$$

where $r = \text{tr}(\Sigma)/\|\Sigma\|$ is the stable rank of $\Sigma^{1/2}$.

$$\|\Sigma_m - \Sigma\| \leq CK^4 \left(\sqrt{\frac{r+u}{m}} + \frac{r+u}{m} \right) \|\Sigma\|$$

with probability at least $1 - 2e^{-u}$.

Proposition 8.6 (Additive Johnson-Lindenstrauss Lemma). *Consider a set $\mathcal{X} \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then, with high probability (say, 0.99), the scaled matrix*

$$Q := \frac{1}{\sqrt{m}} A$$

satisfies

$$\|x - y\|_2 - \delta \leq \|Qx - Qy\|_2 \leq \|x - y\|_2 + \delta \quad \text{for all } x, y \in \mathcal{X}$$

where

$$\delta = \frac{CK^2 w(\mathcal{X})}{\sqrt{m}}$$

and $K = \max_i \|A_i\|_{\psi_2}$.

Theorem 8.7 (M^* bound). *Consider a set $T \subset \mathbb{R}^n$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. Then the random subspace $E = \ker A$ satisfies*

$$\mathbb{E} \text{diam}(T \cap E) \leq \frac{CK^2 w(T)}{\sqrt{m}}$$

where $K = \max_i \|A_i\|_{\psi_2}$.

Corollary 8.8 (Affine sections).

$$\mathbb{E} \max_{z \in \mathbb{R}^n} \text{diam}(T \cap E_z) \leq \frac{CK^2 w(T)}{\sqrt{m}}$$

where $E_z = z + \ker A$.

Theorem 8.9 (Escape theorem). *Consider a set $T \subset S^{n-1}$. Let A be an $m \times n$ matrix whose rows A_i are independent, isotropic and sub-gaussian random vectors in $\mathbb{R}^n$. If*

$$m \geq CK^4 w(T)^2$$

then the random subspace $E = \ker A$ satisfies

$$T \cap E = \emptyset$$

with probability at least $1 - 2 \exp(-cm/K^4)$. Here $K = \max_i \|A_i\|_{\psi_2}$.

9 Chapter 9: Sparse Recovery

High dimension probability in Data science

Chapter 9: Sparse Recovery

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Theorem 9.1. Truth $y = Ax$, $x \in T$, optimization find $x' : y = Ax'$, $x' \in T$. Suppose the rows A_i of A are independent, isotropic and subgaussian random vectors. Then any solution $\hat{x}$ of the optimization satisfies

$$\mathbb{E} \|\hat{x} - x\|_2 \leq \frac{CK^2 w(T)}{\sqrt{m}}$$

where $K = \max_i \|A_i\|_{\psi_2}$

Corollary 9.2 (Sparse recovery: guarantees). Consider optimization

$$\text{Find } x' : y = Ax', \quad \|x'\|_1 \leq \sqrt{s}$$

Assume the unknown s -sparse signal $x \in \mathbb{R}^n$ satisfies $\|x\|_2 \leq 1$. Then x can be approximately recovered from the random measurement vector $y = Ax$ by a solution $\hat{x}$ of the optimization. The recovery error satisfies

$$\mathbb{E} \|\hat{x} - x\|_2 \leq CK^2 \sqrt{\frac{s \log n}{m}}$$

Theorem 9.3 (Exact sparse recovery). Consider optimization program

$$\text{minimize } \|x'\|_1 \quad \text{s.t. } y = Ax' \tag{1}$$

Suppose the rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Then the following happens with probability at least $1 - 2 \exp(-cm/K^4)$.

Assume an unknown signal $x \in \mathbb{R}^n$ is s -sparse and the number of measurements m satisfies

$$m \geq CK^4 s \log n$$

Then a solution $\hat{x}$ of the program is exact, i.e.

$$\hat{x} = x.$$

Definition 9.4 (RIP). An $m \times n$ matrix A satisfies the restricted isometry property (RIP) with parameters α, β and s if the inequality

$$\alpha \|v\|_2 \leq \|Av\|_2 \leq \beta \|v\|_2$$

holds for all vectors $v \in \mathbb{R}^n$ such that $\|v\|_0 \leq s$

Theorem 9.5 (RIP implies exact recovery). Suppose an $m \times n$ matrix A satisfies RIP with some parameters α, β and $(1 + \lambda)s$, where $\lambda > (\beta/\alpha)^2$. Then every s -sparse vector $x \in \mathbb{R}^n$ can be recovered exactly by solving the program (1), i.e. the solution satisfies

$$\hat{x} = x.$$

Theorem 9.6 (Random matrices satisfy RIP). Consider an $m \times n$ matrix A whose rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Assume that

$$m \geq CK^4 s \log(en/s)$$

Then, with probability at least $1 - 2 \exp(-cm/K^4)$, the random matrix A satisfies RIP with parameters $\alpha = 0.9\sqrt{m}, \beta = 1.1\sqrt{m}$ and s .

Theorem 9.7 (Performance of Lasso). *Linear regression setting $Y = X\beta + w$ Consider the Lasso program*

$$\text{minimize } \|y - Ax'\|_2 \text{ s.t. } \|x'\|_1 \leq R \quad (2)$$

Suppose the rows A_i of A are independent, isotropic and sub-gaussian random vectors, and let $K := \max_i \|A_i\|_{\psi_2}$. Then the following happens with probability at least $1 - 2\exp(-s \log n)$.

Assume an unknown signal $x \in \mathbb{R}^n$ is s -sparse and the number of measurements m satisfies

$$m \geq CK^4 s \log n$$

Then a solution $\hat{x}$ of the program (2) with $R := \|x\|_1$ is accurate, namely

$$\|\hat{x} - x\|_2 \leq C\sigma \sqrt{\frac{s \log n}{m}},$$

where $\sigma = \|w\|_{L_2}/\sqrt{m}$.

10 Chapter 10: Dvoretzky-Milman's Theorem

High dimension probability in Data science

Chapter 10: Dvoretzky-Milman's Theorem

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Definition 10.1. Let V be a vector space. A function $f : V \rightarrow \mathbb{R}$ is called *positive-homogeneous* if $f(\alpha x) = \alpha f(x)$ for all $\alpha \geq 0$ and $x \in V$.

The function f is called *subadditive* if

$$f(x + y) \leq f(x) + f(y) \text{ for all } x, y \in V$$

Theorem 10.2 (General matrix deviation inequality). Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries. Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be a positive-homogeneous and subadditive function, and let $b \in \mathbb{R}$ be such that

$$f(x) \leq b\|x\|_2 \quad \text{for all } x \in \mathbb{R}^n$$

Then for any subset $T \subset \mathbb{R}^m$, we have

$$\mathbb{E} \sup_{x \in T} |f(Ax) - \mathbb{E}f(Ax)| \leq Cb\gamma(T)$$

Here $\gamma(T)$ is the Gaussian complexity.

Lemma 10.3 (Sub-gaussian increments). Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, and let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be a positive homogenous and subadditive function satisfying (11.3). Then the random process

$$X_x := f(Ax) - \mathbb{E}f(Ax)$$

has sub-gaussian increments with respect to the Euclidean norm, namely

$$\|X_x - X_y\|_{\psi_2} \leq Cb\|x - y\|_2 \quad \text{for all } x, y \in \mathbb{R}^n.$$

Corollary 10.4 (Johnson-Lindenstrauss Lemma for ℓ_1 norm). Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$, let A be an $m \times n$ Gaussian matrix with i.i.d. $N(0, 1)$ entries, and let $\varepsilon \in (0, 1)$. Suppose that

$$m \geq C(\varepsilon) \log N$$

With high probability the matrix $Q := \sqrt{\pi/2} \cdot m^{-1}A$ satisfies

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_1 \leq (1 + \varepsilon)\|x - y\|_2 \text{ for all } x, y \in \mathcal{X}$$

Corollary 10.5 (Johnson-Lindenstrauss Lemma for ℓ_∞ norm). Let $\mathcal{X}$ be a set of N points in $\mathbb{R}^n$, let A be an $m \times n$ Gaussian matrix with i.i.d. $N(0, 1)$ entries, and let $\varepsilon \in (0, 1)$. Suppose that

$$m \geq N^{C(\varepsilon)}$$

With high probability the matrix $Q := \sqrt{\pi/2} \cdot m^{-1}A$ satisfies

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_1 \leq (1 + \varepsilon)\|x - y\|_2 \text{ for all } x, y \in \mathcal{X}$$

With high probability the matrix $Q := C(\log m)^{-1/2}A$, for some appropriate constant C , satisfies

$$(1 - \varepsilon)\|x - y\|_2 \leq \|Qx - Qy\|_\infty \leq (1 + \varepsilon)\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}$$

Note that in this case $m \geq N$, so Q gives an almost isometric embedding (rather than a projection) of the set $\mathcal{X}$ into ℓ_∞ .

Theorem 10.6 (General Chevet's inequality). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries. Let $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$ be arbitrary bounded sets. Then*

$$\mathbb{E} \sup_{x \in T} \left| \sup_{y \in S} \langle Ax, y \rangle - w(S) \|x\|_2 \right| \leq C \gamma(T) \text{rad}(S)$$

Theorem 10.7 (Random projections of sets). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, and $T \subset \mathbb{R}^n$ be a bounded set. Then the following holds with probability at least 0.99 :*

$$r_- B_2^m \subset \text{conv}(AT) \subset r_+ B_2^m$$

where

$$r_{\pm} := w(T) \pm C \sqrt{m} \text{rad}(T)$$

Theorem 10.8 (voretzky-Milman's theorem: Gaussian form). *Let A be an $m \times n$ Gaussian random matrix with i.i.d. $N(0, 1)$ entries, $T \subset \mathbb{R}^n$ be a bounded set, and let $\varepsilon \in (0, 1)$. Suppose*

$$m \leq c \varepsilon^2 d(T)$$

where $d(T)$ is the stable dimension of T introduced in Section 7.6. Then with probability at least 0.99, we have

$$(1 - \varepsilon)B \subset \text{conv}(AT) \subset (1 + \varepsilon)B$$

where B is a Euclidean ball with radius $w(T)$.

Corollary 10.9 (Gaussian cloud). *Consider a Gaussian cloud of n points in $\mathbb{R}^m$, which is formed by i.i.d. random vectors $g_1, \dots, g_n \sim N(0, I_m)$. Suppose that*

$$n \geq \exp(Cm)$$

with large enough absolute constant C . Show that with high probability, the convex hull the Gaussian cloud is approximately a Euclidean ball with radius $\sim \sqrt{\log n}$.

A random projection of a set T in $\mathbb{R}^n$ onto an m dimensional subspace approximately preserves the geometry of T if $m \gtrsim d(T)$. For smaller m , the projected set PT becomes approximately a round ball of diameter $\sim w_s(T)$, and its size does not shrink with m .

11 My own summary and notes

Summary and notes